

Introduction & Background

Buildings account for ~11% of all global embodied carbon emission (World Green Building Council, n.d.). This includes in large part transitional shelters, as a major contributor (Kuittinen et al., 2015).



Figure 1 . Example of pre-built transitional housing (King, 2022).

Prefabricated disaster relief shelters tend to have high embodied carbon primarily from transporting them—the global market size for which reached \$3.1 billion USD in 2024 (Growth Market Reports, n.d.). An emerging alternative to prefabricated shelters involves *Climate and Local Skill-Centered Design (CLCD)*, integrating digital architectural techniques and locally available materials.

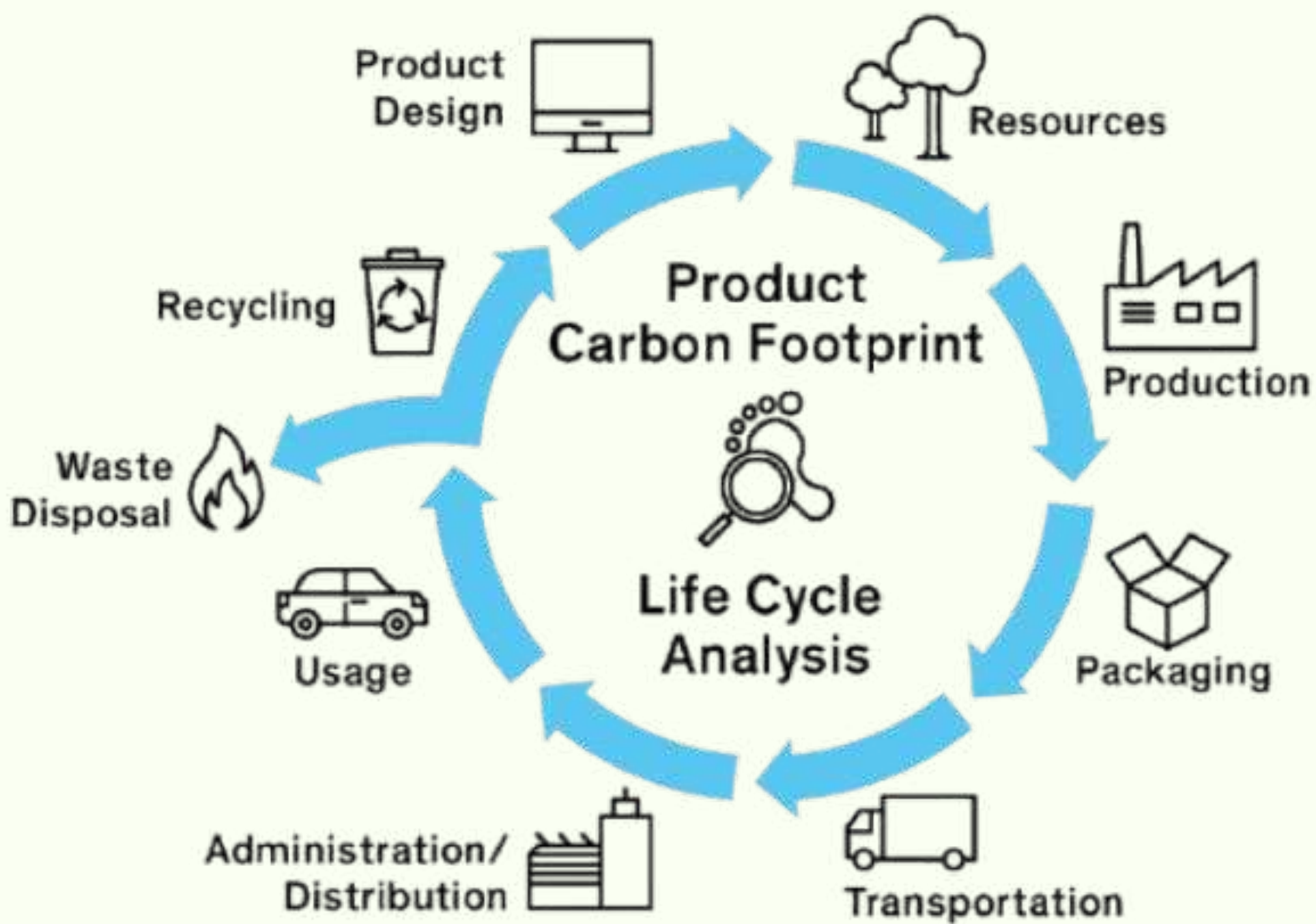


Figure 2 . All considerations in carbon LCA.

Life Cycle Assessment (LCA) calculation tools and analysis are used to holistically determine the embodied carbon output from a production process.

Purpose & Hypothesis

By applying *CLCD* principles, the life-cycle embodied carbon emissions of post-flood transitional housing in the Bangladesh Haor flood plain region will be reduced, lessening the perpetuation of climate change.

Implementing a Digital Design Framework to Reduce the Embodied Carbon of Transitional Humanitarian Shelters

Tools & Methodology

The CLCD process is an entirely digitally facilitated process. Autodesk Revit is the industry standard for architectural Computer-Aided Design (CAD) and Building Information Modeling (BIM), which is a holistic process for creating and managing information about a built asset. It involves the generation and management of digital representations of the physical and functional characteristics of buildings.

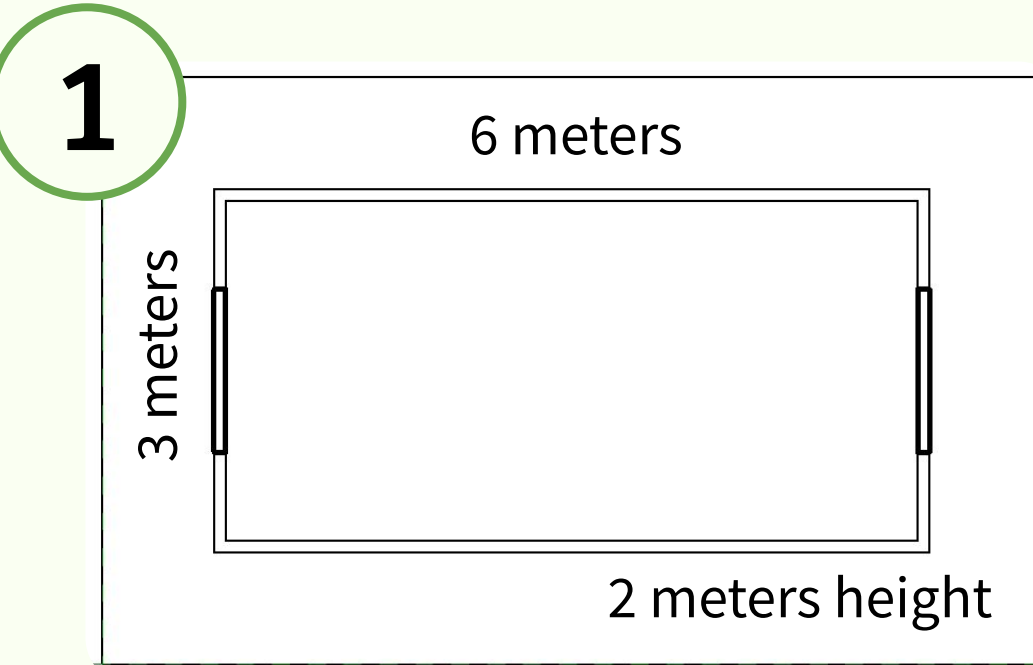


Figure 3 . CAD floor plan.

Make design decisions and complete modeling in Revit to account for 18m² of interior space, fit for ~5-6 people.

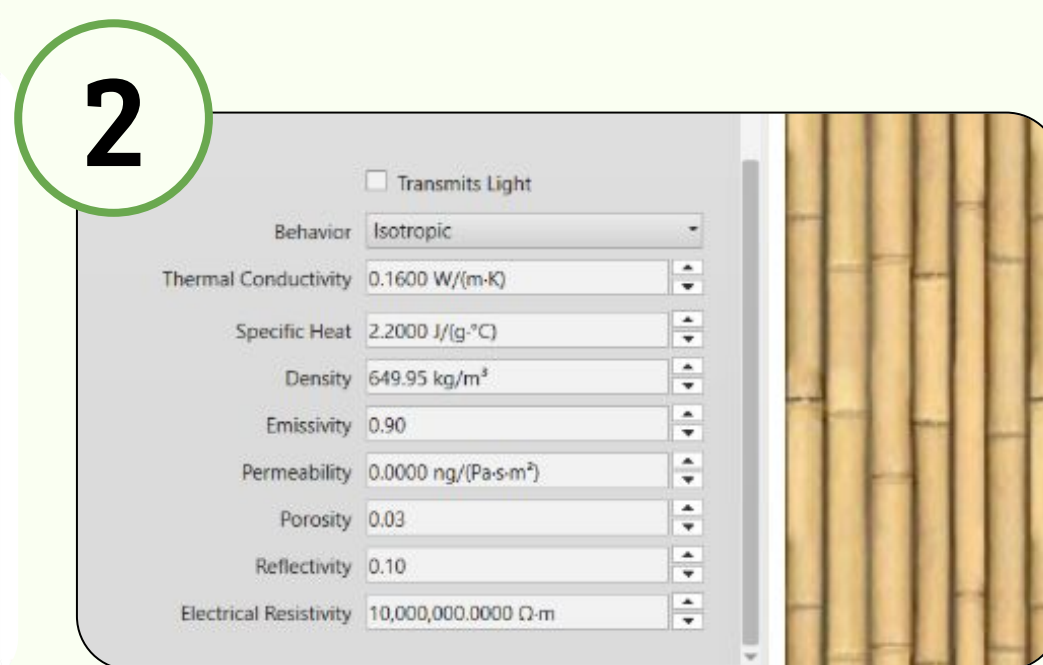


Figure 5 . Bamboo Properties.

Determine material property values and create custom material with texturing to render onto final model.

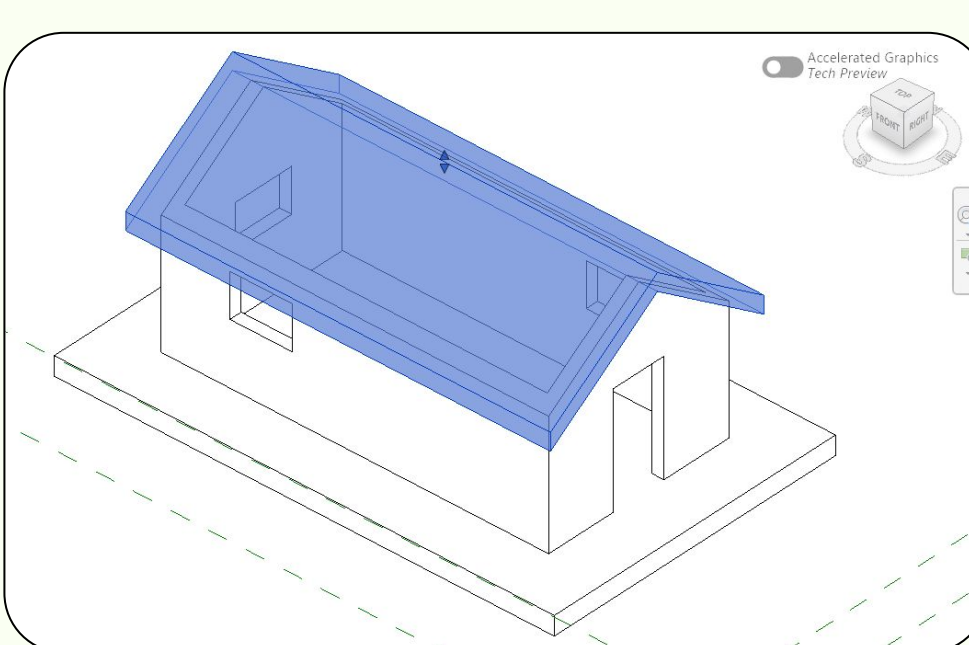


Figure 4 . Final design with unapplied material properties.

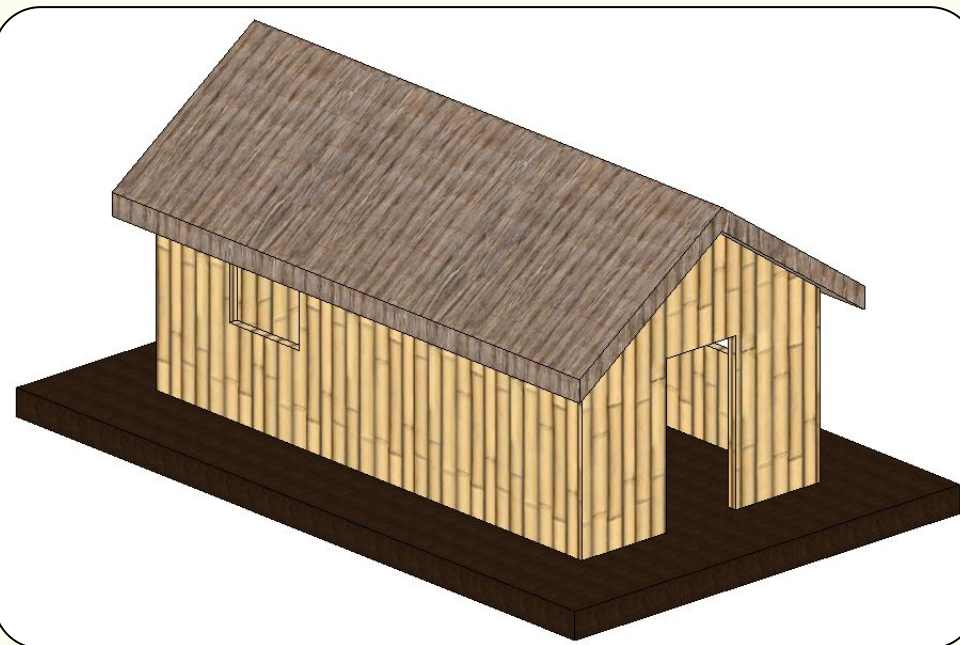


Figure 6 . CAD model with all materials applied.

Using real flood-event time, date, and location, perform daylighting simulations.

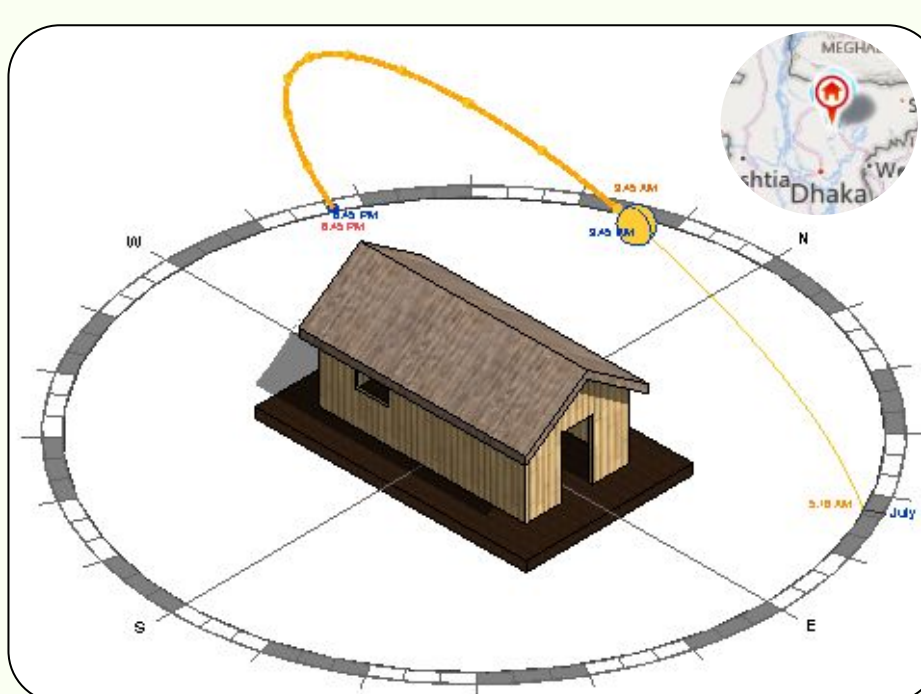


Figure 7 . Daylighting Simulation.

Use daylighting simulation to complete thermal analysis for shelter viability and thermal comfort.

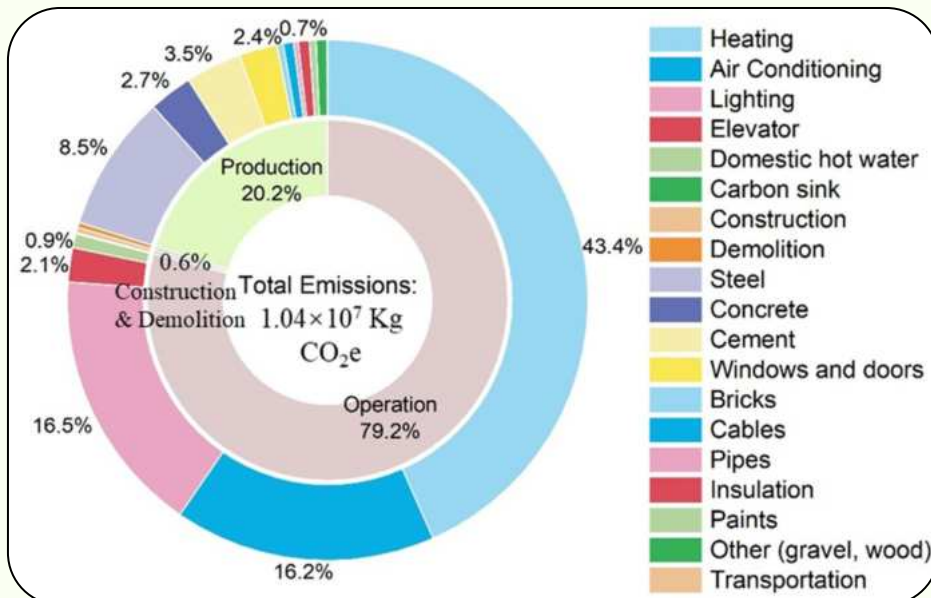
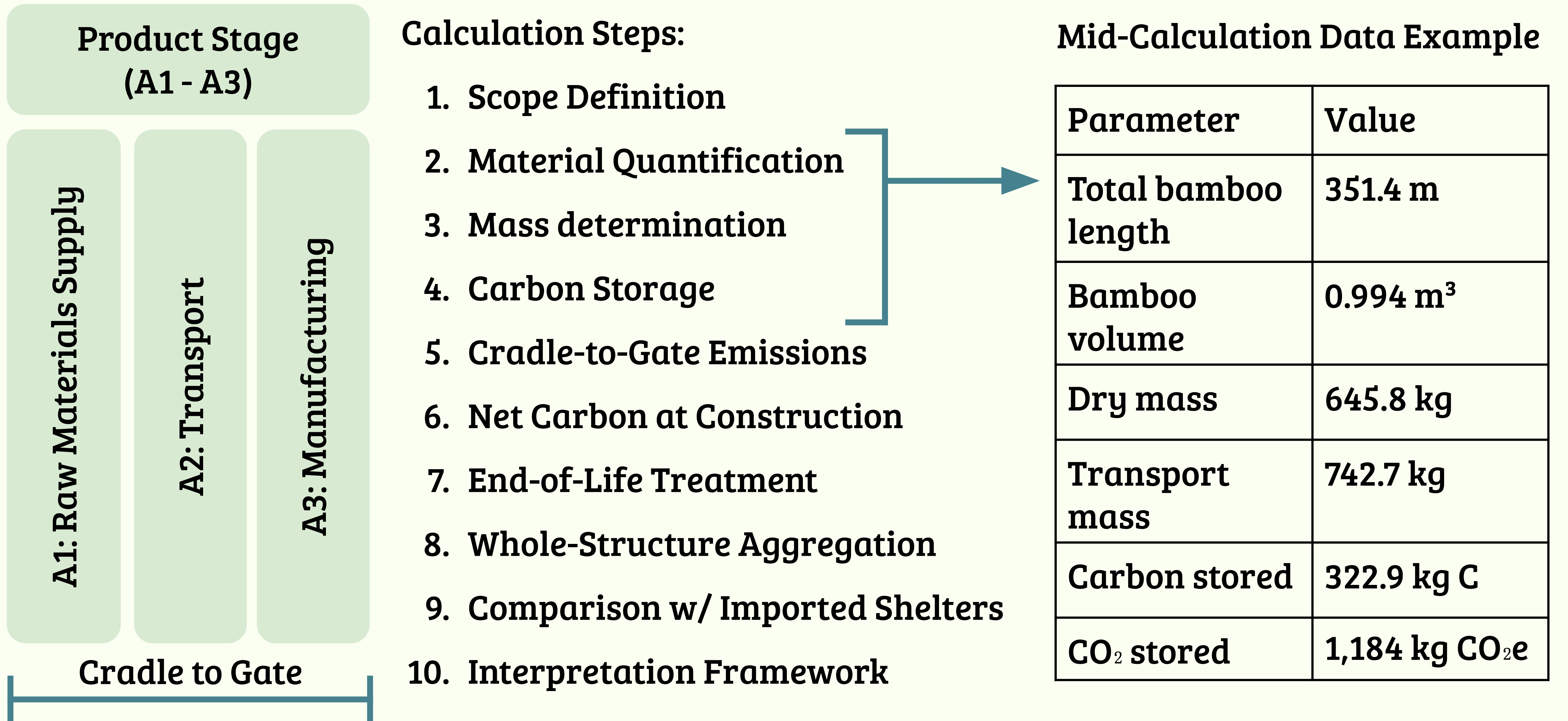


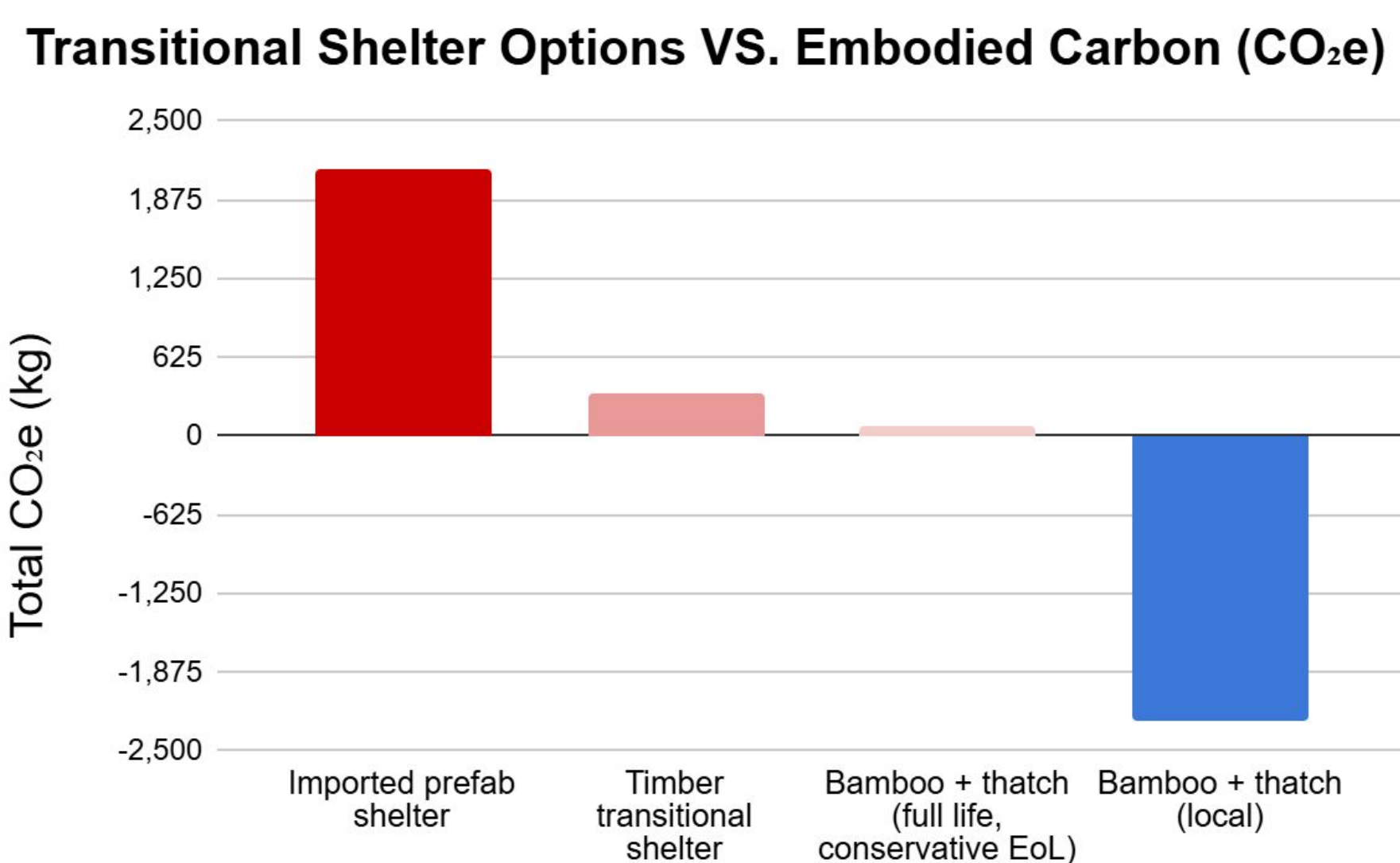
Figure 8 . Example of embodied carbon breakdown.

Using mass information tooling in Revit, complete carbon life-cycle analysis and calculations. Consider emissions related to materials, transportation, and usage.

Calculations



Results



Final *LCA* data compared typical imported transitional shelters with the bamboo structure’s embodied carbon—which was modeled using *two-scenario analysis: full carbon release and full revalorization*. Ultimately, the data shows that imported prefabricated shelters are have high embodied carbon, while the *CLCD* bamboo shelter was a *strong carbon sink* at construction, nearing *net-neutral* in poor end-of-life circumstances.

Conclusion & Extension

The results demonstrate that *CLCD* can significantly *reduce* the embodied carbon of transitional shelters. The *locally sourced* bamboo and thatch structure functioned as a *strong carbon sink* at construction and approached net-neutral emissions under conservative end-of-life conditions. These results support the hypothesis and validate *CLCD* as an effective, low-carbon design framework for post-disaster housing.

Future work could apply *CLCD* to *different geographic case studies*, refine assumptions through *real-world construction* and monitoring, and *collaborate with humanitarian organizations* to implement these strategies at scale.

Citations

Gladdys, M. M., Gentil, B., & Cao, P. (2025). Sustainable Strategies for Improving Humanitarian Construction Through BIM and Climate Analysis. Sustainability, 17(4), 1556. <https://doi.org/10.3390/su17041556>
Growth Market Reports(n.d.) Prefabricated Disaster Relief Shelter Market Research Report 2033. Dataintelio. <https://growthmarketreports.com/report/prefabricated-disaster-relief-shelter-market/amp>
World Green Building Council (n.d.). Bringing embodied carbon upfront. <https://worldgbc.org/climate-action/embodied-carbon/>